

Architectural Performance Analysis of GF(2^{128}) Multipliers

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February 16, 2026

Objective

Finite field multiplication over GF(2^{128}) forms the computational core of modern cryptographic algorithms such as Galois/Counter Mode (GCM) used in secure communication systems. The performance and efficiency of the multiplier directly influence overall system throughput, silicon area, and power consumption in hardware accelerators.

The objective of this work is to systematically evaluate architectural trade-offs between three implementations of a GF(2^{128}) multiplier: (i) a naive polynomial multiplier, (ii) a combinational Karatsuba-based multiplier, and (iii) an internally pipelined Karatsuba architecture.

This study aims to quantify improvements in critical path delay, maximum operating frequency, area utilization, dynamic power, and performance density using the SAED 14nm RVT standard cell library. Particular emphasis is placed on determining whether microarchitectural pipelining provides substantial frequency scaling benefits beyond algorithmic optimization alone.

Architecture Comparison Results

Table 1 summarizes the synthesis results for the three multiplier architectures.

Discussion

The Karatsuba architecture reduces area by approximately 19% compared to the naive implementation while slightly improving timing. However, the most significant improvement is achieved through internal pipelining of the GF(2^{64}) multipliers, which reduces

Table 1: Comparison of GF(2^{128}) Multiplier Architectures (SAED 14nm RVT)

Metric	Naive	Karatsuba (Comb)	Karatsuba + Pipeline	% Change vs Naive
Total Area	33080	26747	26756	↓ 19%
Cell Area	10943	8209	9579	↓ 12%
Critical Path (ns)	~4.2	~4.0	2.23	↓ 47%
Fmax (MHz)	238	250	448	↑ 88%
Latency (cycles)	1	1	3	+2 cycles
Throughput (Mops/s)	238	250	448	↑ 88%
Dynamic Power (mW)	3.864	3.156	7.816	↑ 102%
Leakage (μ W)	425	315	315	↓ 26%
Energy / Operation (mW/MHz)	0.0162	0.0126	0.0174	↑ 7%
Performance / Area	0.00719	0.00934	0.0167	↑ 132%
Performance / Watt	61.6	79.2	57.3	↓ 7%

the critical path from approximately 4.2 ns to 2.23 ns. This results in an 88% increase in maximum operating frequency with negligible area overhead. Although dynamic power increases due to higher switching activity, the performance density (Performance/Area) improves by more than 130%, demonstrating the effectiveness of microarchitectural optimization.